

CS560 - UI/UX Design & Implementation

6-1 Project One: Interactive Application Prototyping and Iterative Design

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Reading Tracker Application Design and Implementation

The goal of this project was to design and implement a responsive Reading Tracker application that allows users to monitor books, track reading progress, and maintain journal entries related to their reading experiences. The application was developed using Vue.js, TypeScript, and Vuetify while applying user experience and interface design principles focused on usability, accessibility, and responsive interaction. My primary objective was to create an application that felt simple and intuitive while still providing meaningful functionality. Many reading applications contain numerous features but often find it difficult to navigate. To address this challenge, I focused on creating a clean dashboard layout, clear navigation paths, and efficient workflows that allow users to manage their reading progress with minimal effort. Another important goal was maintainability. During the design process, I intentionally planned reusable interface elements that could later be implemented as reusable Vue components. This approach helped maintain visual consistency throughout the application while reducing duplicated code. Accessibility and responsive design were also incorporated from the beginning to ensure the application remained usable across desktop, tablet, and mobile devices. Designing applications that are both usable and accessible is an important aspect of modern software development because users expect consistent experiences across multiple devices and platforms (Garofalo, 2025). The result is a Reading Tracker application that combines user-centered design principles with maintainable implementation practices to support an efficient and engaging reading experience.

UX/UI Principles and Design Decisions

Several UX/UI principles influenced the design and implementation of the Reading Tracker application, including usability, accessibility, visual hierarchy, responsive design, and user feedback. Rather than treating these as general design concepts, I applied them to address specific challenges within the application. Usability was a major priority because users need to quickly access information and update their reading progress. I chose a card-based dashboard layout because it allows users to scan information efficiently while keeping content organized. Each book card displays only the most valuable information, including the title, author, page count, and reading progress. Early wireframes included additional statistics, but these created visual clutter and reduced readability. Simplifying the cards created a cleaner and more focused user experience. The dashboard was intentionally designed using a card-based structure because users process grouped visual information more efficiently when content is organized into clear sections with consistent hierarchy (Garofalo, 2025). Visual hierarchy was used to guide attention throughout the interface. Book titles are larger than supporting information, while progress indicators are positioned below the primary content to help users quickly identify reading status. I also chose visual progress bars instead of percentage-only indicators because users can recognize progress more quickly through visual cues than numerical values (Norman, 2013). Accessibility was incorporated throughout the project by adding ARIA labels to icon-based controls and ensuring feedback messages were clearly communicated through snackbar notifications. These decisions help make the application more understandable for users relying on assistive technologies (Garofalo, 2025). Responsive design was implemented using Vuetify's grid system rather than creating separate layouts for different devices. This allowed the interface to adapt automatically across mobile, tablet, and desktop screens while maintaining consistency. Responsive layouts were prioritized because usability can quickly decline when interfaces fail to

adjust appropriately to different screen sizes (Norman, 2013). Finally, immediate feedback was integrated throughout the application. Reading progress updates automatically after users add, edit, or delete entries, and confirmation messages provide reassurance that actions were completed successfully. Providing clear feedback improves visibility of system status and strengthens user confidence in the application (Nielsen, 2020).

User Types, Goals, and Constraints

The Reading Tracker application was designed around two primary user groups whose goals and constraints influenced both interface organization and interaction design. The first user type is a casual reader who uses the application for personal reading enjoyment. This user primarily wants a straightforward way to track progress and occasionally record reading notes. Their goals include reviewing reading status, identifying completed books, and maintaining a lightweight reading history. The biggest challenge for this user is simplicity. If the application becomes overly complex or requires excessive navigation, it may discourage continued use. To support this user group, I minimized navigation depth and kept the dashboard focused on essential information. The second user type is an academic or goal-oriented reader. This user manages multiple books simultaneously and often records detailed reading notes. Their priorities include accurate progress tracking, maintaining reading history, and documenting thoughts related to specific sections of a book. Efficiency becomes particularly important for this user because they may update entries frequently. To support these needs, I designed modal-based editing workflows that allow users to update information without leaving the current page context. These user groups influenced several design decisions throughout the application. For example, the Add Entry modal supports quick updates for casual readers while still providing enough flexibility for users who want detailed journal entries. Similarly, the responsive dashboard

ensures that both user groups can access their reading information across multiple devices without sacrificing usability.

Workflow Architecture and Navigation Structure

The workflow architecture of the Reading Tracker application was intentionally designed to minimize navigation complexity and keep users focused on reading-related activities. During the planning phase, I considered creating separate pages for adding, editing, and deleting journal entries. While technically feasible, that approach would require users to navigate through multiple screens for simple actions. Instead, I organized the application around two primary screens: the Book Dashboard and the Book Detail page. The dashboard provides a high-level overview of all books and reading progress. Selecting a book opens the Book Detail page, where users can manage journal entries and update progress. This separation allows the dashboard to remain clean while providing a dedicated space for detailed reading management. Reducing unnecessary navigation supports usability because users can focus on completing tasks rather than learning complicated interface structures (Nielsen, 2020). I intentionally centralized management features within the Book Detail page because distributing them across multiple screens would create unnecessary workflow interruptions. One of the most important workflow decisions involved in using modal dialogs for journal management is the ability to manage. Rather than redirecting users to separate pages, modal overlays allow users to add or edit entries while remaining connected to the current book. Keeping users within the same workflow context minimizes interruptions and reduces cognitive switching costs (Norman, 2013). The workflow also incorporates immediate feedback mechanisms. After saving or deleting an entry, the application automatically updates progress information and displays confirmation messages. This

design reinforces confidence that actions have been successfully completed and contributes to a smoother overall experience.

Low-Fidelity Wireframes and Interactive Prototype Design

The low-fidelity wireframes for the Reading Tracker application were created in Figma to establish the structure, navigation, and user flow before implementation began. Rather than focusing on visual styling, I concentrated on layout organization and information hierarchy. This allowed me to evaluate usability decisions early and adjust before development. One of the most important design decisions involved simplifying the dashboard. Early wireframes included additional reading statistics and management controls, but these features created visual clutter and made it harder to quickly identify books and reading progress. After reviewing the layouts, I reduced the amount of information displayed and focused on the most important content. This created a cleaner and more readable interface. Establishing a clear visual hierarchy during the wireframing stage helped users identify essential information more efficiently and improved navigation predictability (Garofalo, 2025). The interactive prototype expanded the wireframes by introducing realistic user interactions. Users can browse books, view detailed reading information, and manage journal entries through clickable navigation flows. During prototype testing, I discovered that navigating to a separate page for adding entries interrupted the user experience. Replacing that workflow with a modal dialog allowed users to remain on the Book Detail page while updating progress, creating a smoother and more efficient experience. Keeping users within the same workflow context minimizes interruptions and reduces cognitive switching costs (Norman, 2013). The prototype also helped validate responsive behavior across desktops, tablets, and mobile layouts. Testing different screen sizes revealed spacing and alignment issues

that could be corrected before implementation. Identifying these usability concerns about early reduced development rework and supporting an iterative design process (Preece et al., 2019).

Interface Implementation and Engineering Decisions

The implementation phase translated the Figma prototype into a functional Vue.js and TypeScript application. While the prototype focused on usability and navigation, implementation required decisions related to maintainability, scalability, accessibility, and component organization. One of my primary goals was maintaining consistency between the prototype and the final application. To achieve this, I relied on reusable components. The BookCard component was reused throughout the application rather than creating multiple versions of similar layouts. This reduced duplicated code while ensuring that typography, spacing, progress indicators, and interactions remained consistent. Reusable component structures improve maintainability while also promoting consistency across the user interface (Garofalo, 2025). Another crucial decision involved using Vuetify's responsive grid system to support mobile, tablet, and desktop layouts. Instead of creating separate interfaces for each device, the layout automatically adjusts based on screen size. This approach reduced code duplication while preserving responsive behavior. Separating layout behavior from content logic improved scalability and supported responsive functionality across multiple device sizes (Norman, 2013). Accessibility was also integrated during implementation. ARIA labels were added to icon-based controls, and snack bar notifications provide clear feedback when users complete actions such as adding, editing, or deleting entries. These improvements help make the application more accessible while improving usability for all users. I also reused the same modal component for both adding and

editing journal entries. This simplified maintenance, reduced duplicated code, and ensured a consistent experience throughout the application.

Interactive Prototype

Figma Interactive Prototype:

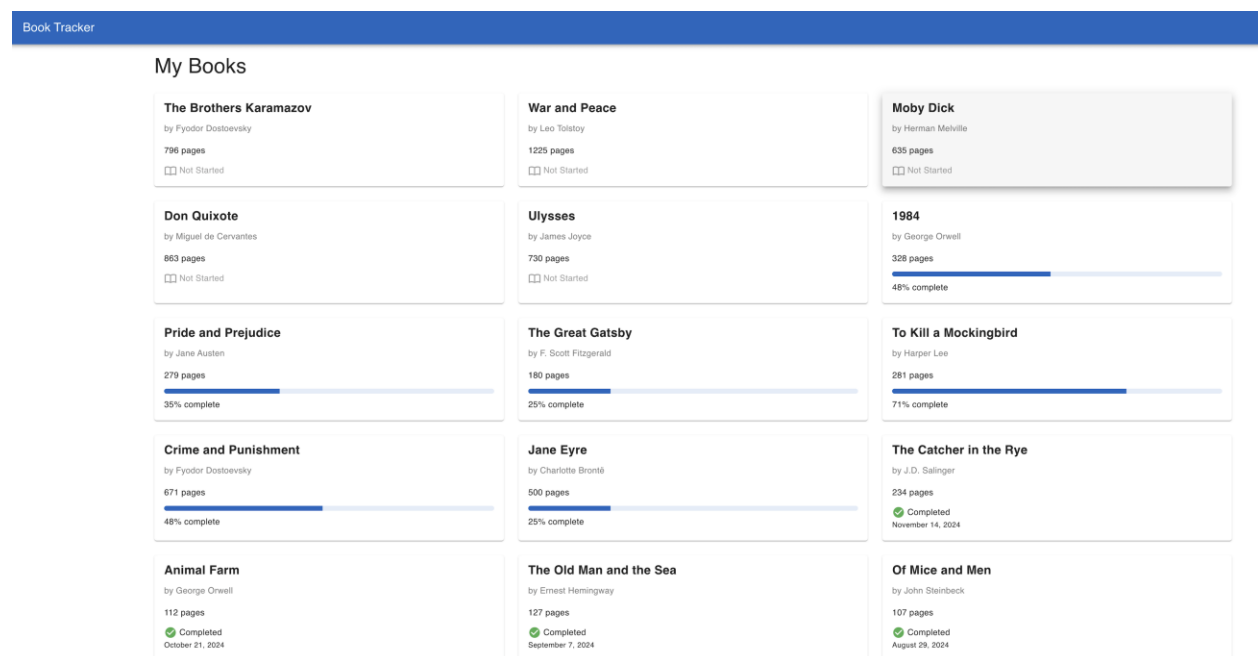
The interactive prototype demonstrates the core user flows of the Reading Tracker application, including viewing books, navigating to book details, creating journal entries, editing books, and interacting with responsive layouts across desktop, tablet, and mobile devices.

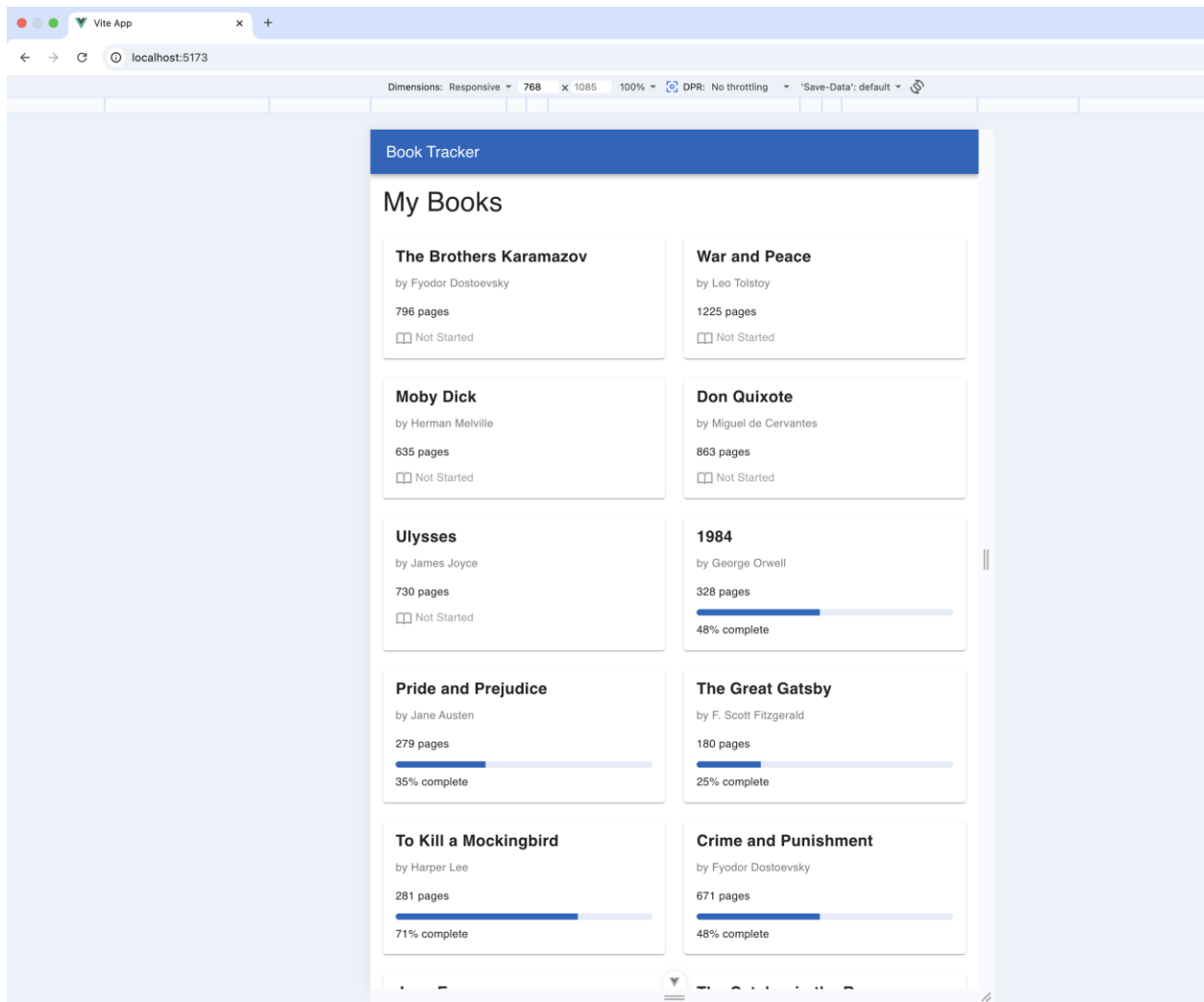
<https://www.figma.com/proto/N85GgbYCCW0KWOHXpMO7To/Book-Tracker--M-Debska?node-id=2110-3160&p=f&t=eTCrFzYRM90aLzFw-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=2068%3A285>

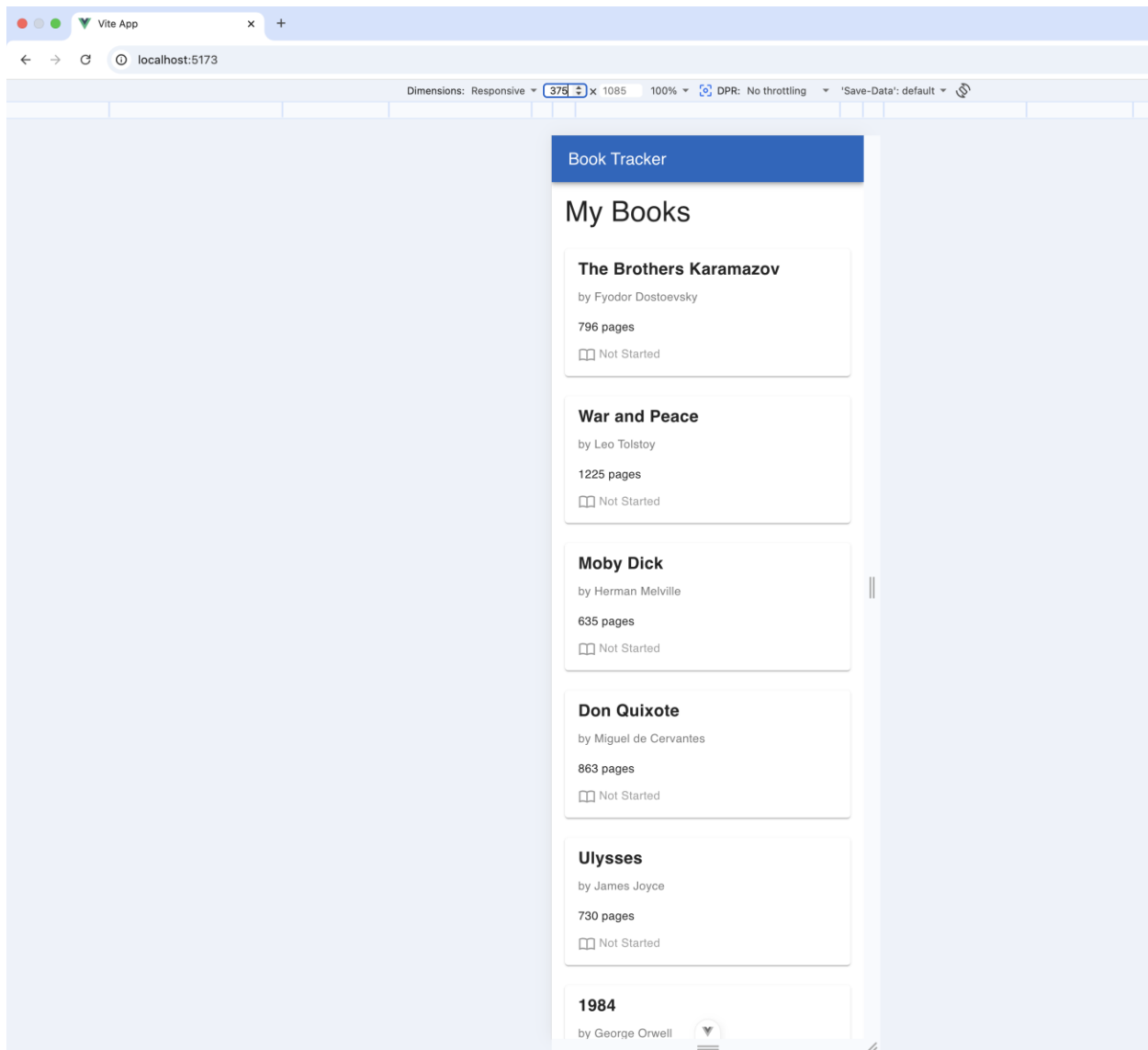
One of the most valuable lessons I learned during this project was that useful design decisions often involve balancing functionality with simplicity. Early versions of the dashboard included additional controls and information because I wanted users to access every feature quickly. However, testing showed that too much information reduced readability and distracted users from the primary task of tracking reading progress. Simplifying the dashboard improved usability while still preserving functionality. I also gained a better understanding of how prototyping supports software development. The Figma prototype was not only useful for planning the interface but also helped identify workflow issues before implementation began. Several design improvements, including modal-based journal entry management and responsive layout adjustments, were made during prototype testing rather than during coding. Another

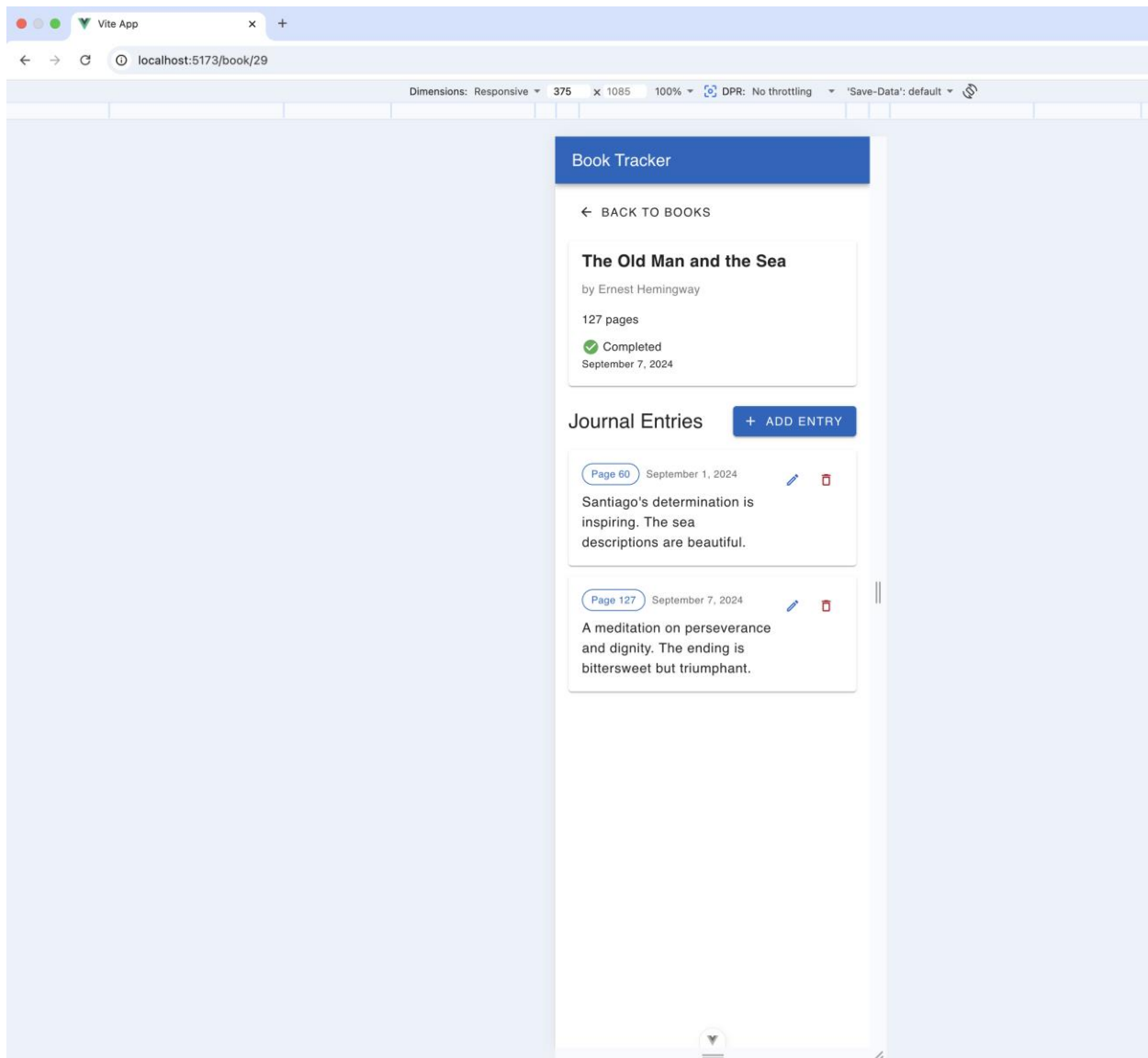
important takeaway was the value of reusable components. Reusing the same BookCard and modal structures improved consistency across the application while reducing maintenance effort. This project reinforced the importance of considering maintainability and scalability during the design phase rather than waiting until implementation is complete. If I continue developing the Reading Tracker application, I will add search functionality, personalized reading goals, advanced filtering, and reading analytics. However, I would maintain the simplified workflow because keeping the application intuitive and easy to navigate became one of the most successful outcomes of this project.

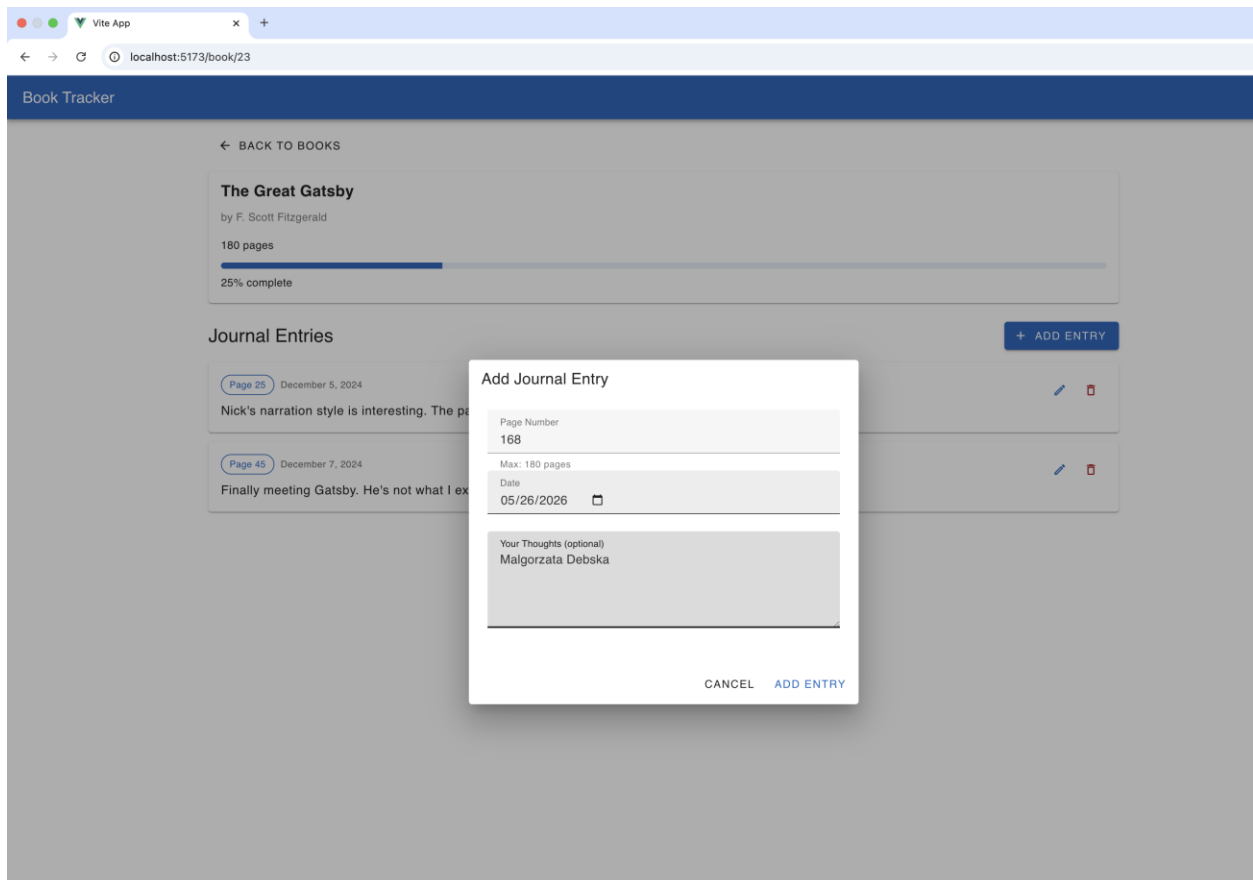
No AI tools were used for the completion of this assignment.





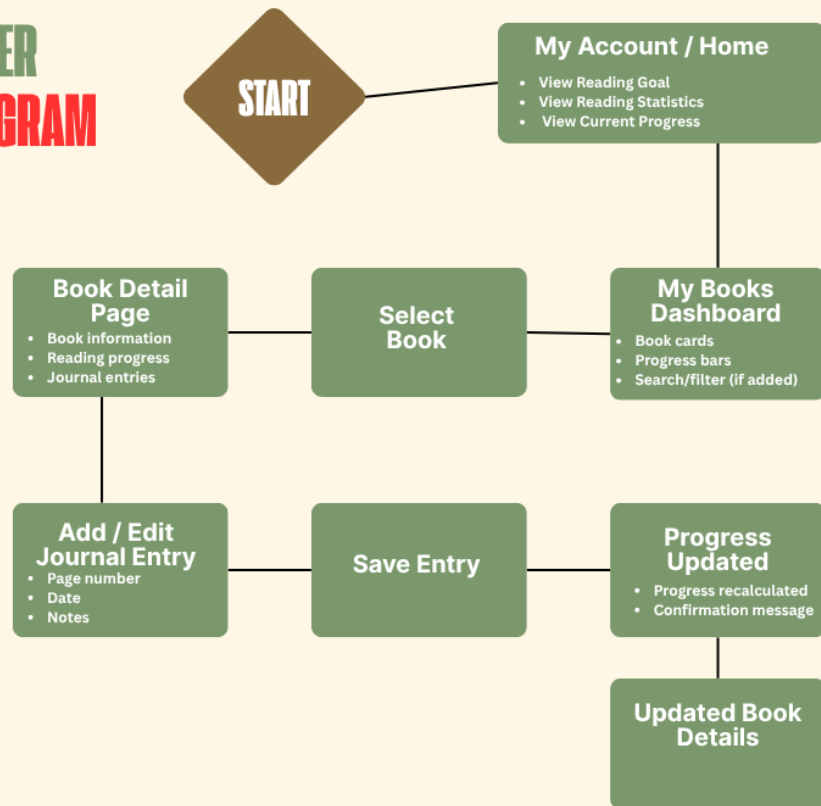






Workflow Diagram:

READING TRACKER WORKFLOW DIAGRAM



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